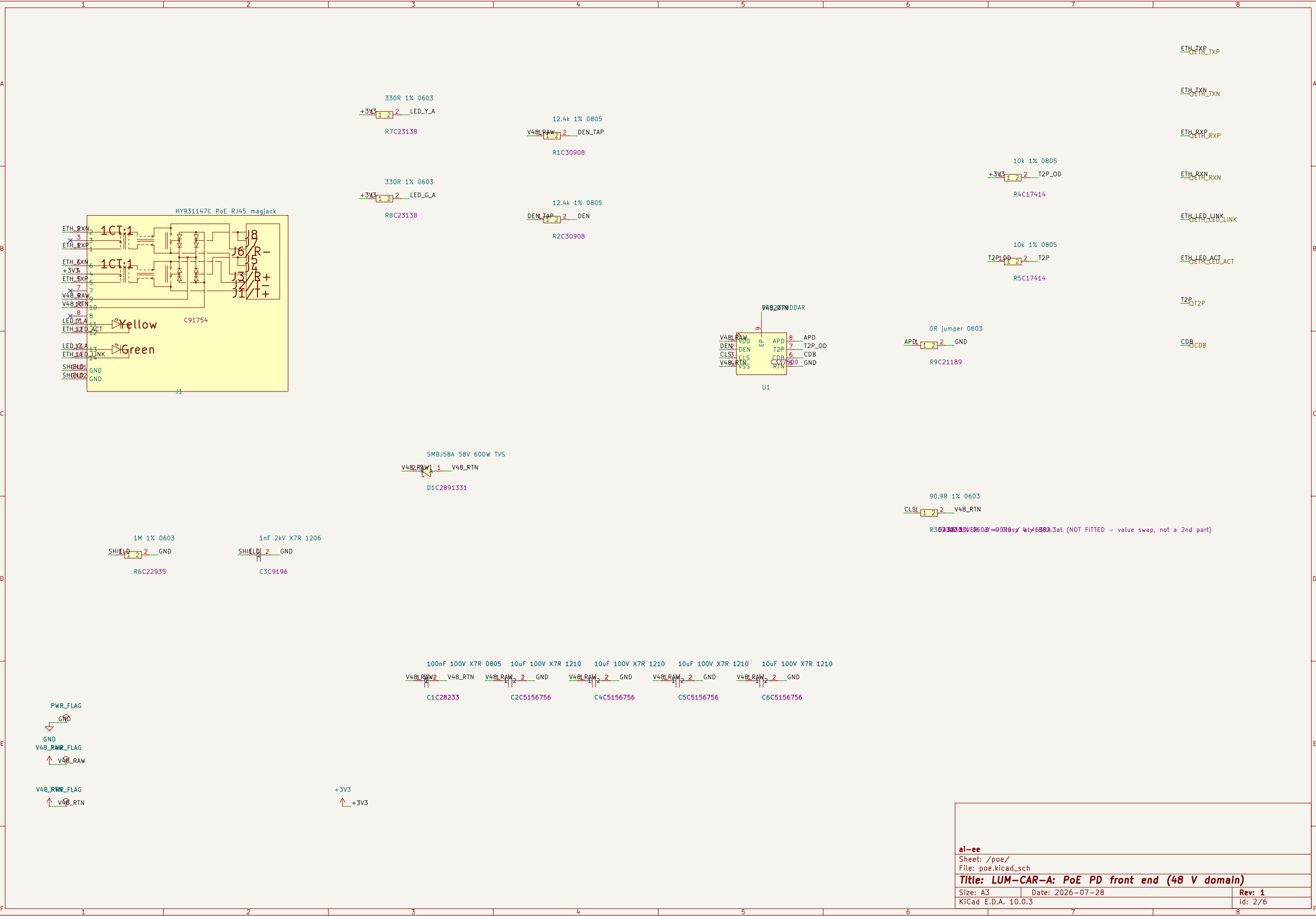
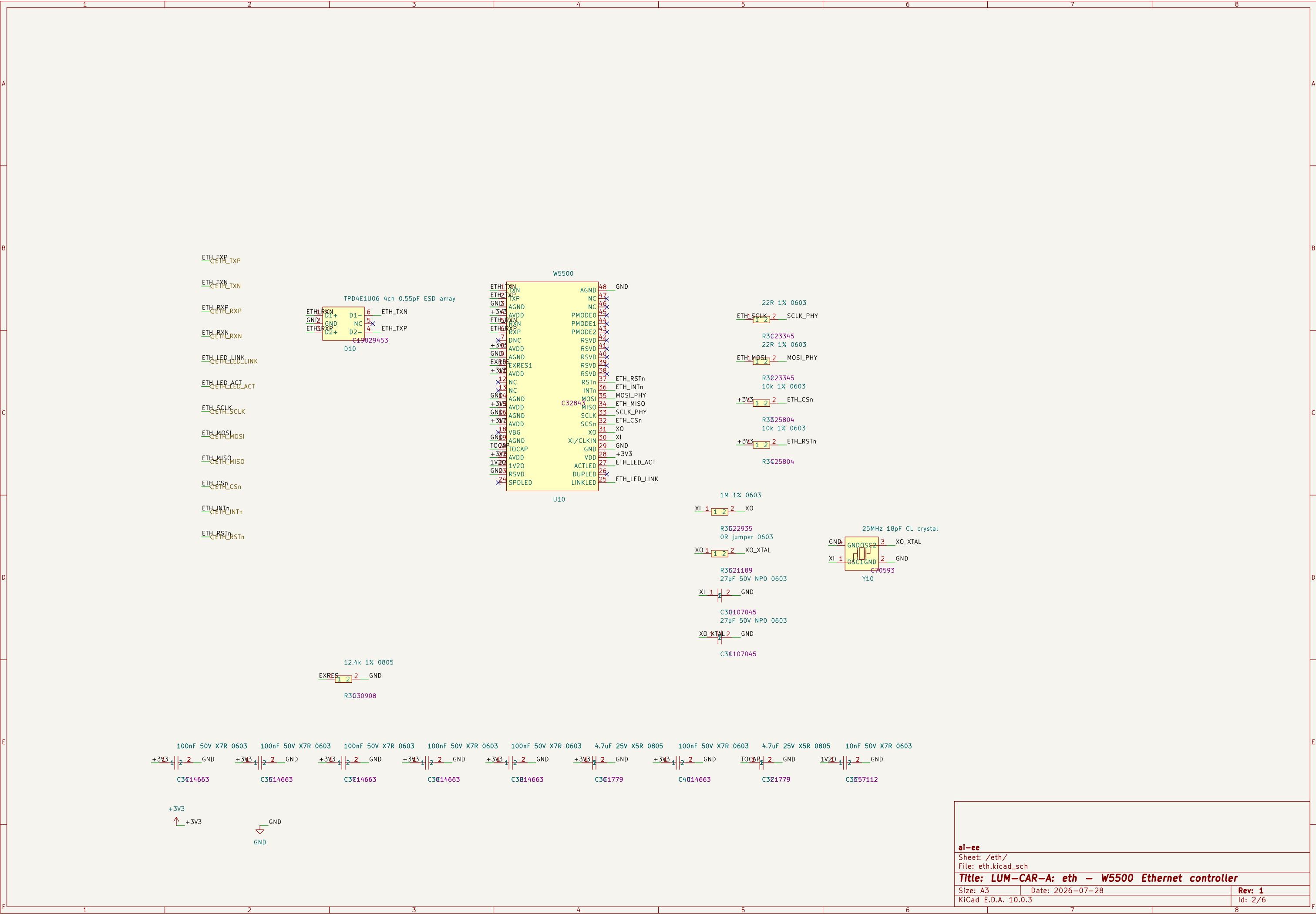
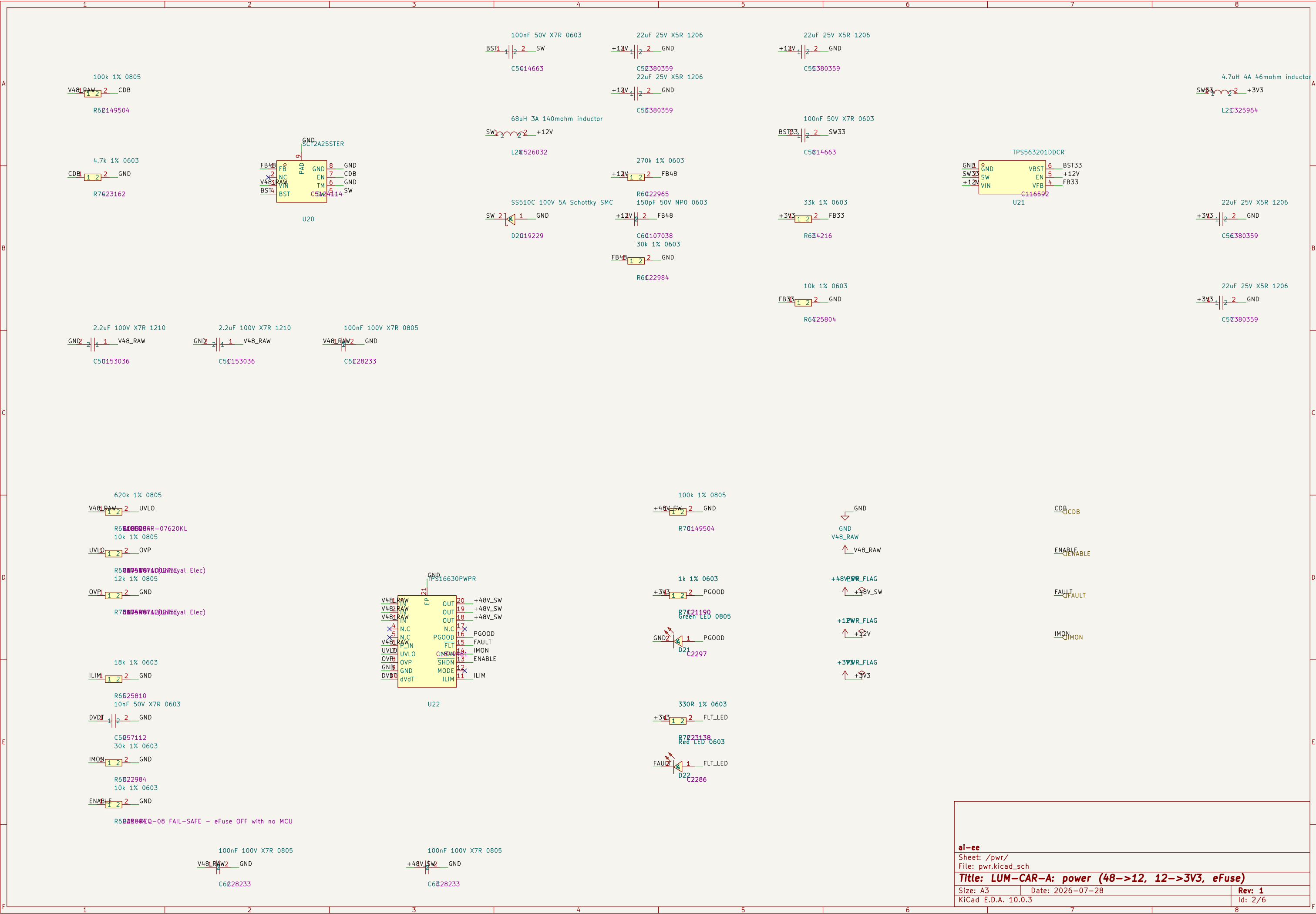


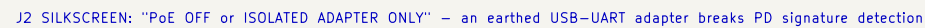
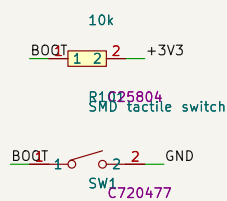
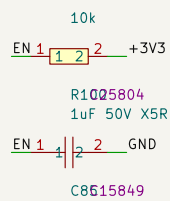








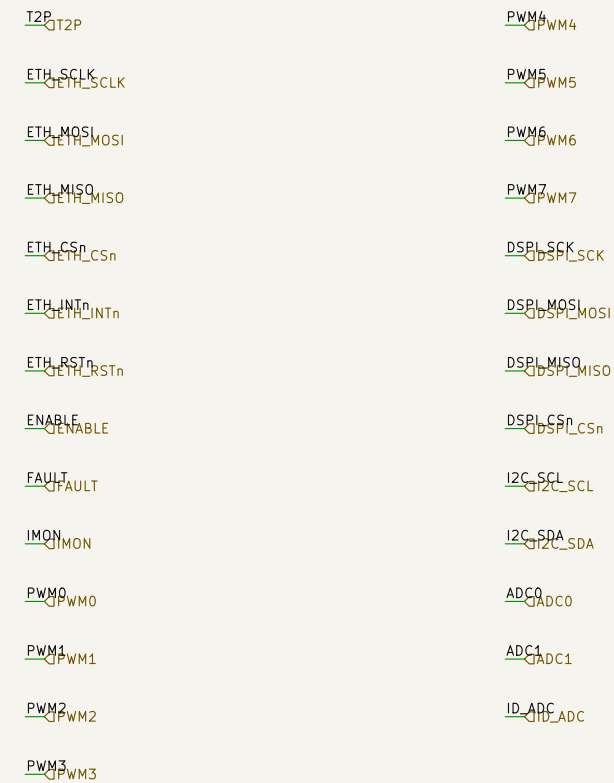
|                                                  |                  |         |
|--------------------------------------------------|------------------|---------|
| ai-ee                                            |                  |         |
| Sheet: /pwr/<br>File: pwr.kicad_sch              |                  |         |
| Title: LUM-CAR-A: power (48->12, 12->3V3, eFuse) |                  |         |
| Size: A3                                         | Date: 2026-07-28 | Rev: 1  |
| KiCad E.D.A. 10.0.3                              |                  | Id: 2/6 |



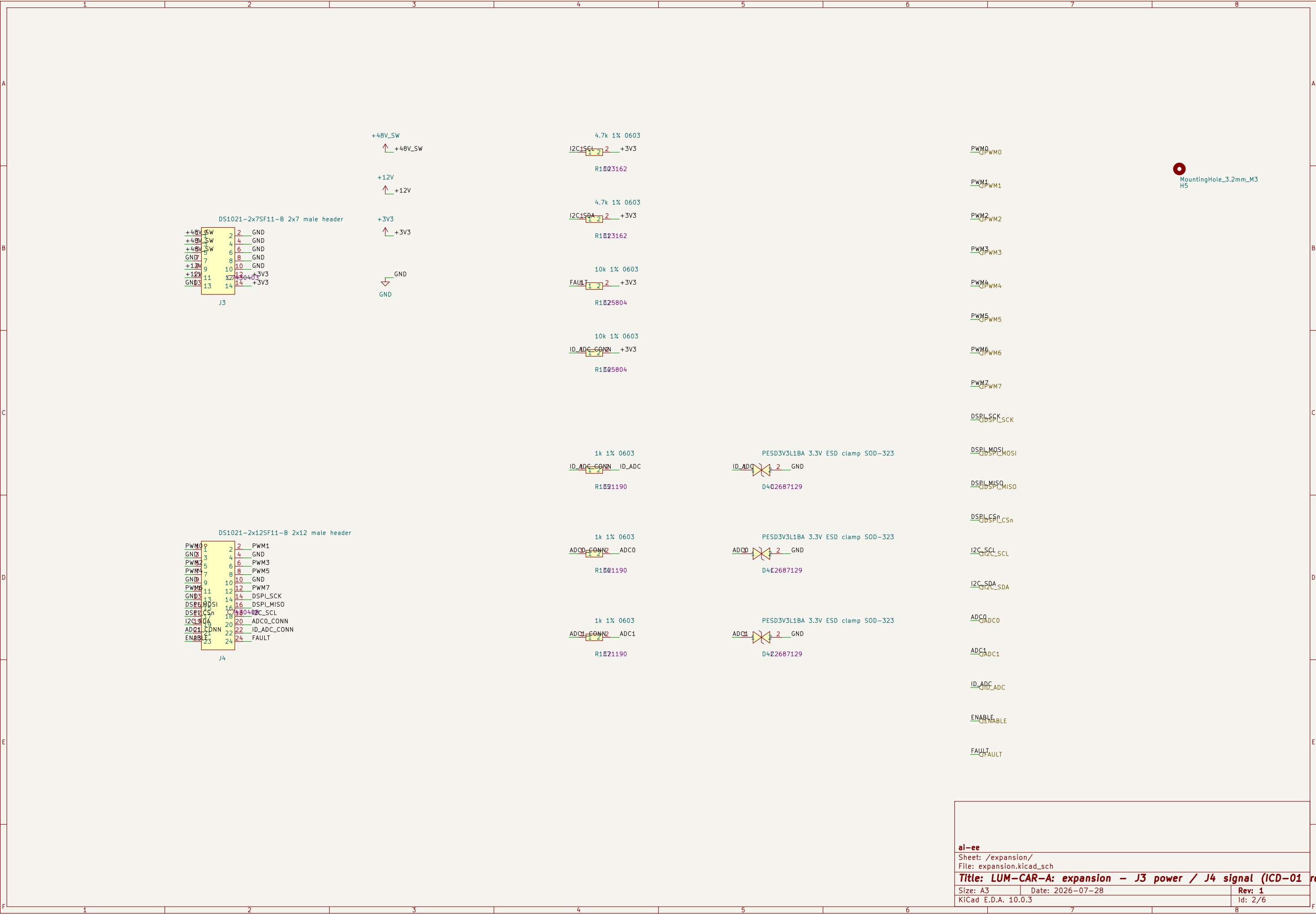
GPIO45 (U30 pad 26) IS A NO-CONNECT BY DESIGN: pulled high at power-up on this no-PSRAM -N8 it selects 1.8 V VDD\_SPI and the 3.3 V flash never boots

BOOT idles HIGH through R101 (GPIO0 = 1 -> SPI Boot); SW1 and J2 pin 6 can only pull it low

Keep the antenna end of U30 (the pad 1 / pad 40 corner) clear of copper, parts and traces on EVERY layer







|                                                                    |                  |         |
|--------------------------------------------------------------------|------------------|---------|
| ai-ee                                                              |                  |         |
| Sheet: /expansion/<br>File: expansion.kicad_sch                    |                  |         |
| Title: LUM-CAR-A: expansion - J3 power / J4 signal (ICD-01 rev A2) |                  |         |
| Size: A3                                                           | Date: 2026-07-28 | Rev: 1  |
| KiCad E.D.A. 10.0.3                                                |                  | Id: 2/6 |